Percentage Increases & Decreases

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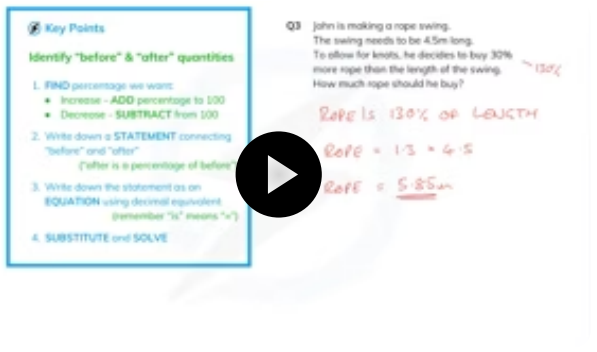

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Percentage increases & decreases

How do I increase by a percentage?

- A percentage **increase** makes an amount **bigger** by **adding** that percentage on to itself
- **Without a calculator**, use the the **basic percentages** methods to find the percentage you are increasing by
  - Then **add** this on to the original amount
  - To increase 30 by 10%
    - 10% of 30 is 3
    - 30 + 3 = 33
    - This is equivalent to finding 110% of 30
- **With a calculator** it is more efficient to use **multipliers**
  - A multiplier is the **decimal equivalent** of a percentage
    - A **percentage** can be converted **to a decimal** by **dividing by 100**
  - When **increasing** by a percentage, we are finding a percentage **greater than 100%**
  - To increase 80 by 15%
    - We are finding **115%** of 80, so the multiplier is **1.15**
    - 1.15 × 80 = 92

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- A percentage **decrease** makes an amount **smaller** by **subtracting** that percentage from itself
- **Without a calculator**, use the methods outlined in **Basic Percentages** to find the percentage you are decreasing by
  - Then subtract this from the original amount
  - To decrease 30 by 10%
    - 10% of 30 is 3
    - 30 - 3 = 27
    - This is equivalent to finding 90% of 30
      - Because 100% - 10% = 90%

- **With a calculator** it is more efficient to use **multipliers**
  - When **decreasing** by a percentage, we are finding a percentage **smaller than 100%**
  - To decrease 80 by 15%
    - We are finding **85%** of 80, so the multiplier is **0.85**
      - Because 100% - 15% = 85%

    - 0.85 × 80 = 68

Worked Example

(a) Increase 200 kg by 21%.

Answer:

Method 1 (non-calculator)

By first finding 10% and 1%, find 21% of 200

10% of 200 = 20
1% of 200 = 2
21% of 200 = 20 + 20 + 2 = 42

Add this to the original amount

200 + 42

242 kg

Method 2 (calculator)

An increase by 21% is equivalent to finding 121% of the original amount
So the multiplier is 1.21

1.21 × 200

242 kg

(b) An item that costs £500 is discounted by 35%.

Find the new price of the item.

Answer:

A discount of 35% means the price decreases by 35%

10% of 500 = 50
5% of 500 = 25
35% of 500 = 50 + 50 + 50 + 25 = 175

Subtract this from the original amount

500 - 175

£325

Method 2 (calculator)

A decrease of 35% is equivalent to finding 65% of the original amount (100 - 35 = 65)
So the multiplier is 0.65

500 × 0.65

£325

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